

HS22 Barcode Scanner

User Manual



Revision

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Table of Content

Chapter 1 System Setting	11
Introduction	11
Programming Barcode	11
Use of programming barcode	11
Serial Command Programming	12
Restore to factory default	12
Read Firmware Version	12
User Preference	12
Beeper	12
Programming Beep	12
Beeper Frequency	13
Startup Beep	13
Good Read Beep	14
Error Beep	14
Illumination LED	15
Aiming LED	16
Indicator LED	16
Sleep Mode	17
Timer for enter sleep mode	17
Chapter 2 Scan Mode	18
Scan Mode	18

Auto-sensing Mode	18
Level Mode(default) (Trigger)	18
Continuous Mode	18
Command Mode	20
Command Mode Scanning Setting	20
Decode Session Timeout	20
Only valid in sense mode, level trigger mode and command mode.	20
Reread Timeout	21
Chapter 3 Communication Interface	22
Interface	22
USB Keyboard (default)	22
USB CDC	22
RS-232	23
HID POS	23
USB Keyboard	23
USB Countries Keyboard	23
USB Transmit Speed	25
Control Character (ASCII Character Conversion)	26
Polling Rate	26
Serial Interface	27
Baud Rate	27
Stop Bit	28

Parity Check	28
Chapter 4 Data Format	29
Encoding Format	29
Input Encoding Format	29
Output Encoding Format	30
Prefix/Suffix Sequence	30
Prefix	31
Custom Prefix	31
Suffix	32
Custom Suffix	32
Code ID	33
Code ID On/Off	33
Terminator Suffix	34
Convert Case	34
Chapter5 Symbologies	36
Introduction	36
Enable/Disable All Symbologies	36
Enable/Disable ALL 1D Symbologies	37
Enable/Disable All 2D Symbologies	37
Inverse Barcode	37
1D Inverse Barcode	37
2D Inverse Barcode	38

EAN-13	38
Enable/Disable	38
Check Digit	38
Addenda 2 digit	38
Addenda 5 Digit	39
Addenda Required	39
ISSN	39
Enable/Disable	39
2 Digit Addenda	39
Addenda Require	40
ISBN	40
Enable/Disable	40
5 Digit Addenda	40
Addenda Required	40
EAN-8	41
Enable/Disable	41
Check Digit	41
Convert EAN8 to EAN13	41
2 Digit Addenda	41
5 Digit Addenda	42
Addenda Required	42
UPC-A	42

Enable/Disable	42
Check Digit	42
Convert UPC-A to EAN13	43
2 Digit Addenda	43
5 Digit Addenda	43
Addenda Require	43
UPC-A System Character	44
UPC-E	44
Enable/Disable	44
Check Digit	44
Convert UPC-E to UPC-A	45
2 Digit Addenda	45
5 Digit Addenda	45
Addenda Require	45
Code 128	46
Enable/Disable Code 128	46
Set Length Range to Code128	46
Code 39	47
Enable/Disable Code39	47
Check Digit	47
Start/Stop Character	47
Code 39 Full ASCII	48

Set Length Range to Code39.....	48
Code 32	48
Code 93	49
Enable/Disable	49
Check Digit.....	49
Set Length Range to Code93.....	50
Codabar	50
Enable/Disable	50
Start/Stop Character	51
Check Digit.....	51
Set Length Range for Codabar	51
Interleaved 2 of 5 (ITF5)	52
Enable/Disable	52
Check Digit.....	52
Let Length Range for ITF5.....	53
Industrial 2 of 5	54
Enable/Disable	54
Check Digit.....	54
Set the length range for Industrial 2 of 5	55
Matrix 2 of 5	55
Enable/Disable	56
Check Digit.....	56

Set Length Range for Matrix 2 of 5	56
Code 11	57
Enable/Disable	57
Check Digit	57
Set Length Range for Code11	58
MSI-Plessey	58
Enable/Disable	58
Check Digit	59
Set Length Range for MSI-Plessey	60
GS1-DataBar (RSS)	60
RSS-14 Enable/Disable	60
RSS-Limited Enable/Disable	61
RSS-Expanded Enable/Disable	61
RSS-Stacked Enable/Disable	61
Set Length Range for RSS-Stacked	61
Micro QR Code	62
QR Code	63
Enable/Disable	63
QR code with URL	63
Data Matrix	63
PDF417	63
Micro PDF417	64

Aztec Code	64
Appendix	65
Appendix 1 Digital Barcodes	65
Appendix 2 Symbologies	67
Appendix 3 Serial Command Format	68
Appendix 4 Control Character	69
Appendix 5 ASCII Table	70

Chapter 1 System Setting

Introduction

There are two ways to configure the HS-22 barcode scanner: barcode programming and command programming.

Programming Barcode



The figure above is an example that shows you the programming barcode and command for the Enter Setup function:

1. The **Enter Setup** barcode.
2. The description of feature/option.
3. (default) indicates factory default settings.

Use of programming barcode

Programming Barcode



Enter Setup (default)

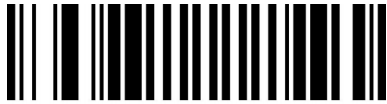


Exit Setup

Serial Command Programming

Please refer to Serial command protocol.

Restore to factory default



Factory Default



Factory Default-Inverse Barcode

Read Firmware Version



Read Firmware

User Preference



Save User Preference

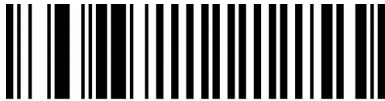


Restore to User Preference

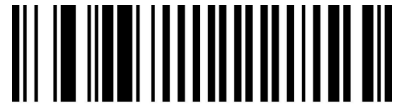
Beeper

Programming Beep

Enable or disable beep for programming barcode successfully.

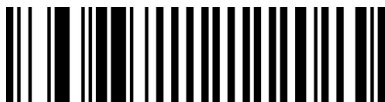


Enable(default)

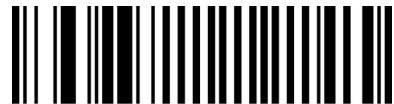


Disable

Command Programming Beep



Enable

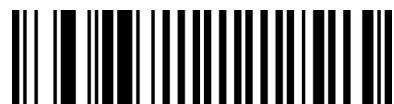


Disable(default)

Beeper Frequency



2.0K



2.4K



2.7K(default)

Startup Beep



Enable(default)



Disable

Good Read Beep



Enable(default)



Disable

Good Read Beep Volume



Highe(35)(default)



Medium(20)



Low(10)



Tips

The default value for good read beep is 35, settable range is from 1 to 50.

Error Beep



Enable(default)



Disable

Illumination LED

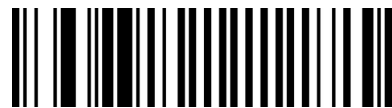
Normal(default) : Illumination LEDs on the scanner are turned on during image capture.

Always On: Illumination LEDs on the scanner keep on after the engine is powered on.

Off: Illumination LEDs on the scanner are off all the time.



Normal(default)



Always On



Off

Aiming LED

When scanning/capturing image, the scanner projects an aiming pattern which allows positioning the target barcode within its field of view and thus makes decoding easier

Normal (default) : The scanner projects an aiming pattern only during barcode scanning.

Always On: Aiming pattern is constantly on after the engine is powered on.

Off: Aiming pattern is off all the time.



Normal(default)



Always On



Off

Indicator LED



Enable(default)



Disable

Sleep Mode



Enable



Disable(default)



Sleep mode is only valid in level trigger mode and command mode, but invalid in auto-sensing mode and continuous mode.

Timer for enter sleep mode

The default value is 1 second, settable range: 1~99 seconds.



1s(default)



5s



10s



Enter Sleep Mode Now

Chapter 2 Scan Mode

Scan Mode

Auto-sensing Mode

The scanner activates a decode session every time it detects a barcode presented to it. The decode session continues until a barcode is decoded or the decode session timeout expires. **Reread Timeout** can avoid undesired rereading of same barcode in a given period of time.



Sense Mode

Level Mode(default) (Trigger)

A trigger pull activates a decode session. The decode session continues until a barcode is decoded or you release the trigger



Level Mode(default)

Continuous Mode

The scanner automatically starts one decode session after another. To suspend/resume barcode reading, simply press the trigger. **Reread Timeout** can avoid undesired rereading of same barcode in a given period of time.



Continuous Mode

Command Mode

The scanner only works with command. (Refer to 《Serial Command Protocol》)

Under the RS-232 t, USB CDC and HID POS interface, the scanner can enter the command mode.



In "command mode", scanner is not allowed to change the communication interface. If you need to modify it, please switch to other reading modes first.

Command Mode Scanning Setting

Decode Session Timeout

Only valid in sense mode, level trigger mode and command mode.



3s(default)



7s



10s

Reread Timeout

Reread Timeout can avoid undesired rereading of same barcode in a given period of time.

This feature is only applicable to the Sense and Continuous modes.



0ms



500ms(default)



1s



Set Custom Timeout

Example

Example: Set reread timeout to 1000ms.

- 1、Scan Enter Setup barcode
- 2、Scan Set Custom Timeout barcode
- 3、Scan numeric barcodes: "1""0""0""0" (Appendix 1)
- 4、Scan Save barcode" (Appendix 1 Save or Cancel barcode)
- 5、Scan Exit Setup Barcode.

Chapter 3 Communication Interface

Interface

USB Keyboard (default)

The scanner's transmission is simulated as USB keyboard input with no need for command configuration or a driver. Barcode data could be entered by the virtual keyboard directly and it is also convenient for the host device to receive data.



USB Keyboard

USB CDC

It is compliant with the standard USB CDC class specifications defined by the USB-IF and allows the host device to receive data in the way as a serial port does. A driver is needed when using this feature



USB CDC

RS-232

Serial communication interface is usually used when connecting the engine to a host device (like PC, POS). However, to ensure smooth communication and accuracy of data, you need to set communication parameters (including baud rate, parity check, data bit and stop bit) to match the host device.



RS-232

HID POS

It is based on the HID interface, with no need for a custom driver. It excels virtual keyboard and traditional TTL-232 interface in transmission speed



HID POS

USB Keyboard

USB Countries Keyboard

Keyboard layouts vary from country to country. The default setting is U.S. keyboard.



U.S. (default)



United Kingdom



Italy



Brazil(Spanish)



Portugal(Portuguese)



Brazil (Portuguese)



French(Franch)



Austria(German)



Swiss(German)



Turkish Q



Turkish F



Japan (Japanese)

USB Transmit Speed



5ms(default)



10ms



20ms

Control Character (ASCII Character Conversion)

◦



Disable(default)



Control + ASCII Mode (Ctrl+ASCII)



Alt + Keypad Mode

Polling Rate

This parameter specifies the polling rate for a USB keyboard. If the Host drops characters, change the polling rate to a bigger value

The default value is 5ms, settable range: 1ms~255ms.



1ms



5ms(default)



10ms

Serial Interface

Baud Rate



Baud Rate 1200



Baud Rate 4800



Baud Rate 9600(default)



Baud Rate 14400



Baud Rate 19200



Baud Rate 38400



Baud Rate 57600



Baud Rate 115200

Stop Bit



Stop bit 1(default)



Stop Bit 2

Parity Check



No Parity Check(default)



Odd Parity



Even Parity

Chapter 4 Data Format

Encoding Format

Input Encoding Format

Auto Discriminate: Decode UTF-8 and GBK barcodes both.

UTF-8: only decode UTF-8 barcodes.

GBK: only decode GBK barcodes.



Auto Discriminate (default)



UTF-8



GBK

Output Encoding Format



Original Data



UTF-8



GBK(default)



Notice: Only for USB CDC, RS-232 and HID POS.

Prefix/Suffix Sequence



Code ID+Prefix+data+Suffix+Terminator

Suffix (default)



Prefix+Code ID+Data+Suffix+Terminator

Prefix

If custom prefix is enabled, you are allowed to append to the data a user-defined prefix that cannot exceed 10 characters.

For example, if the custom prefix is "AB" and the barcode data is "123", the Host will receive "AB123"



Notice: All symbolologies will be same prefix.

Custom Prefix



Enable



Disable(default)



Set Custom Prefix



Set prefix to 'a'(Hex: 0x61)

1.Scan Enter Setup barcode.

2.Scan Set Custom Prefix barcode

3.Scan numeric barcode: "6" "1" (Appendix 1)

4.Scan Save barcode (Appendix 1 Save or Cancel Barcode)

5.Scan Exit Setup barcode

Suffix

If custom suffix is enabled, you are allowed to append to the data a user-defined suffix that cannot exceed 10 characters.

For example, if the custom suffix is "AB" and the barcode data is "123", the Host will receive "123AB"



Notice: The custom suffix will be same for all symbologies.

Custom Suffix



Enable



Disable(default)



Set Custom Suffix

Example

Set Suffix to 'a'(Hex: 0x61)

- 1.Scan Enter Setup barcode.
- 2.Scan Set Custom Suffix barcode
- 3.Scan numeric barcode: "6""1" (Appendix 1)
- 4.Scan Save barcode (Appendix 1 Save or Cancel Barcode)
- 5.Scan Exit Setup barcode

Code ID

Code ID On/Off



Disable CODE ID (default)



Enable CODE ID



Set Custom Code ID

Example

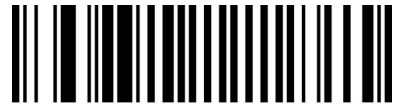
Set Codabar (Code ID:0x42) Code ID to "Y" (Hex: 0x59) :

1. Scan Enter Setup barcode
2. Scan Set Custom Code ID
3. Scan numeric barcodes: "4""2""5""9" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Exit Setup barcode.

Terminator Suffix



CR (Enter) (default)



LF (Down)



CR+LF (Enter+Down)



Tab



ETX



No Terminator Suffix

Convert Case

Case conversion is performed on the characters in the string, without affecting the prefix and suffix.



Normal(default)



Convert Case



Convert all characters to Upper Case



Convert all characters to Lower Case

Chapter5 Symbologies

Introduction

Every symbology (barcode type) has its own unique attributes. This chapter provides programming barcodes for configuring the engine so that it can identify various symbologies. It is recommended to disable those that are rarely used to increase the efficiency of the scanner.

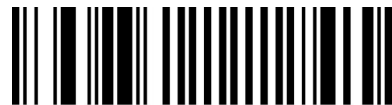


Note: The maximum length limit of any one-dimensional barcode cannot exceed 127.

Enable/Disable All Symbologies



Disable All Symbologies



Enable All Symbologies



Restore default

Enable/Disable ALL 1D Symbologies



Enable All 1D Symbologies



Disable All 1D Symbologies



Restore default

Enable/Disable All 2D Symbologies



Disable All 2D Symbologies



Enable All 2D Symbologies



Restore Default

Inverse Barcode

1D Inverse Barcode



Decode both regular and inverse 1D
barcode



Only Decode Regular 1D Barcode(default)

2D Inverse Barcode



Decode Both Regular and Inverse
Barcode



Decode only Regular Barcode(default)

EAN-13

Enable/Disable



EAN13 Enable(default)



EAN13 Disable

Check Digit



Transmit EAN13 Check Digit (default)



Do not Transmit EAN13 Check Digit

Addenda 2 digit



Decode 2 digit Addenda



Do not Decode 2 Digit Addenda (default)

Addenda 5 Digit



Decode 5 Digit Addenda



Do Not Decode 5 Digit Addenda(default)

Addenda Required



Required



Not Required(default)

ISSN

Enable/Disable



ISSNE Enable



ISSN Disable(default)

2 Digit Addenda



Decode 2 Digit Addenda



Do not Decode 2 Digit Addenda(default)

Addenda Require



Required



Not Required(default)

ISBN

Enable/Disable



ISBN Enable



ISBN Disable(default)

5 Digit Addenda



Decode 5 Digit Addenda



Do not Decode 5 Digit Addenda(default)

Addenda Required



Required



Not Required(default)

EAN-8

Enable/Disable



EAN8Enable(default)



EAN8Disable

Check Digit



Transmit EAN8 Check Digit (default)



Transmit EAN8 Check Digit

Convert EAN8 to EAN13



Convert EAN8 to EAN13



Do not Convert EAN8 to EAN13 (default)

2 Digit Addenda



Decode 2 Digit Addenda



Do not Decode 2 Digit Addenda(default)

5 Digit Addenda



Decode 5 Digit Addenda



Do not Decode 5 Digit Addenda(default)

Addenda Required



Required



Not Required(default)

UPC-A

Enable/Disable



UPC-A Enable(default)



UPC-A Disable

Check Digit



Transmit UPC-A Check Digit (default)



Do not Transmit UPC-A Check Digit

Convert UPC-A to EAN13



Convert UPC-A to EAN13



Do not Convert UPC-A to EAN13(default)

2 Digit Addenda



Decode 2 Digit Addenda



Do not Decode 2 Digit Addenda(default)

5 Digit Addenda



Decode 5 Digit Addenda



Do not Decode 5 Digit Addenda(default)

Addenda Require



Required



Not Required(default)

UPC-A System Character



Output System Character(default)



Do Not Output System Character

UPC-E

Enable/Disable



UPC-E0 Enable(default)



UPC-E0 Disable



UPC-E1 Enable(default)



UPC-E1 Disable

Check Digit



Enable UPC-E Check Digit (default)



Disable UPC-E Check Digit

Convert UPC-E to UPC-A



Convert UPC-E to UPC-A



Do not Convert UPC-E to UPC-A (default)

2 Digit Addenda



Decode 2 Digit Addenda



不 Decode 2 Digit Addenda(default)

5 Digit Addenda



Decode 5 Digit Addenda



不 Decode 5 Digit Addenda(default)

Addenda Require



Required



Not Required(default)

Code 128

Enable/Disable Code 128



Code128 Enable(default)



Code128 Disable

Set Length Range to Code128



Set the Minimum Length



Set the Maximum Length

Example

Set the length range of code128 from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcode "8" (Appendix 1 Digital barcodes)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcode "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Code 39

Enable/Disable Code39



Code39 Enable(default)



Code39 Disable

Check Digit



Disable Code39 Check Digit(default)



Enable Code39 Check Digit, Do not Transmit Check Digit



Enable Code39 Check Digit, Transmit Check Digit

Start/Stop Character



Output Start/Stop Character



Do not Output Start/Stop Character (default)

Code 39 Full ASCII



Enable



Disable(default)

Set Length Range to Code39



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12 :

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Code 32



Code32Enable



Code32Disable(default)

Code 93

Enable/Disable



Code93 Enable(default)



Code93 Disable

Check Digit



Disable Code93 Check Digit



Enable Code93 Check Digit, Do not Transmit Check Digit (default)



Enable Code93 Check Digit, Transmit Check Digit

Set Length Range to Code93



Set the Minimum Length



Set the Maximum Length

Example

Set the scanner to decode Code39 barcode containing between 8 to 12 characters:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Codabar

Enable/Disable



Codabar Enable (default)



Codabar Disable

Start/Stop Character



Transmit Start/Stop Character



Do not Transmit Start/Stop
Character(default)

Check Digit



Disable Codabar Check Digit(default)



Enable Codabar Check Digit, Do not
Transmit Check Digit



Enable Codabar Check Digit, Transmit Check
Digit

Set Length Range for Codabar



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Interleaved 2 of 5 (ITF5)

Enable/Disable



Interleaved 2 of 5 (ITF5) Enable



Interleaved 2 of 5 (ITF5) Disable(default)

Check Digit



Disable Interleaved 2 of 5 (ITF5) Check
Digit(default)



Enable Interleaved 2 of 5 (ITF5) Check Di
git,

Do not Transmit Check Digit



Enable Interleaved 2 of 5 (ITF5) Check Digit,
Transmit Check Digit

Let Length Range for ITF5

ITF-6 is a special kind of Interleaved 2 of 5 with a length of 6 characters and the last character as the check character.

ITF-6 priority principle: For the Interleaved 2 of 5 barcodes with a length of 6 characters and the last character as the check character, the ITF-6 configurations shall take precedence over the Interleaved 2 of 5 settings.

ITF-14 is a special kind of Interleaved 2 of 5 with a length of 14 characters and the last character as the check character.

ITF-14 priority principle: For the Interleaved 2 of 5 barcodes with a length of 14 characters and the last character as the check character, the ITF-14 configurations shall take precedence over the Interleaved 2 of 5 settings.



Set the Minimum Length



Set the Maximum Length



Set to ITF-6



Set to ITF-14

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Industrial 2 of 5

Enable/Disable



Industrial 2 of 5Enable



Industrial 2 of 5Disable(default)

Check Digit



Disable Industrial 2 of 5Check Digit(default)



Enable Industrial 2 of 5 Check Digit, Do not Transmit Check Digit



Enable Industrial 2 of 5 Check Digit, Transmit Check Digit

Set the length range for Industrial 2 of 5



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1" "2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Matrix 2 of 5

Enable/Disable



Matrix 2 of 5 Enable



Matrix 2 of 5 Disable (default)

Check Digit



Disable Matrix 2 of 5 Check Digit (default)



Enable Matrix 2 of 5 Check Digit, Do not
Transmit Check Digit



Enable Matrix 2 of 5 Check Digit, Transmit
Check Digit

Set Length Range for Matrix 2 of 5



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode

-
2. Scan Set the Minimum Length
 3. Scan numeric barcodes "8" (Appendix 1)
 4. Scan Save barcode (Appendix 1)
 5. Scan Set the Maximum Length barcode
 6. Scan numeric barcodes "1""2" (Appendix 1)
 7. Scan Save barcode (Appendix 1)
 8. Scan Exit Setup barcode

Code 11

Enable/Disable



Code 11 Enable



Code 11 Disable(default)

Check Digit



Disable Code 11Check Digit(default)



Enable Code 11Check Digit, Do not Transmit
Check Digit



Enable Code 11Check Digit, Transmit Check
Digit

Set Length Range for Code11



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

MSI-Plessey

Enable/Disable



MSI-Plessey Enable



MSI-Plessey Disable(default)

Check Digit



Disable MSI-Plessey Check Digit



Enable MSI-Plessey One Bit Check
Digit(default)



Enable MSI-Plessey Two Bit Check Digit



MSI-Plessey Transmit Check Digit (default)



MSI-Plessey Do not Transmit Check Digit

Set Length Range for MSI-Plessey



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

GS1-DataBar (RSS)

RSS-14 Enable/Disable



RSS-14 Enable



RSS-14 Disable(default)

RSS-Limited Enable/Disable



RSS-Limited Enable



RSS-Limited Disable(default)

RSS-Expanded Enable/Disable



RSS-Expanded Enable



RSS-Expanded Disable(default)

RSS-Stacked Enable/Disable



RSS-Stacked Enable



RSS-Stacked Disable(default)

Set Length Range for RSS-Stacked



Set the Minimum Length



Set the Maximum Length

Example

Set Code39 length range from 8 to 12:

1. Scan Enter Setup barcode
2. Scan Set the Minimum Length
3. Scan numeric barcodes "8" (Appendix 1)
4. Scan Save barcode (Appendix 1)
5. Scan Set the Maximum Length barcode
6. Scan numeric barcodes "1""2" (Appendix 1)
7. Scan Save barcode (Appendix 1)
8. Scan Exit Setup barcode

Micro QR Code



Micro QR Enable



Micro QR Disable(default)

QR Code

Enable/Disable



QR Enable(default)



QR Disable

QR code with URL



Enable



Disable(default)

Data Matrix



Data Matrix Enable(default)



Data Matrix Disable

PDF417



PDF417 Enable(default)



PDF417 Disable

Micro PDF417



Micro PDF417 Enable



Micro PDF417 Disable(default)

Aztec Code



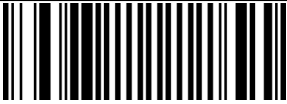
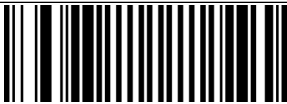
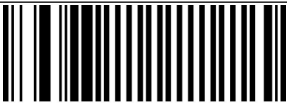
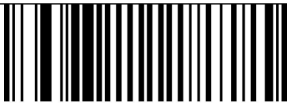
Aztec Code Enable

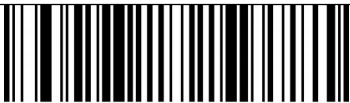
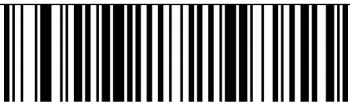
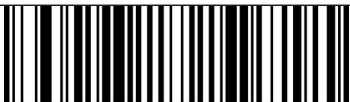
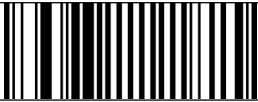


Aztec Code Disable(default)

Appendix

Appendix 1 Digital Barcodes

	
0	
	
	1
	
2	
	
	3
	
4	
	
	5
	
6	
	
	7
	
8	
	
	9

	
A	
	
	B
	
C	
	
	D
	
E	
	
	F
	
Save	
	
	Cancel
	
Delete All Digits	
	
	Delete Last Digit

Appendix 2 Symbolologies

Barcode Type	Code ID	
	HEX	Char
Codabar	0x42	B
Code 11	0x68	h
Code 128	0x6A	j
Code 39(Code 32)	0x62	b
Code 93	0x69	i
ISBN	0x53	S
ISSN	0x6E	n
EAN		
EAN-13(including ISBN)	0x64	d
EAN-8	0x44	D
GS1		
GS1 DataBar	0x79	y
GS1 DataBar Limited	0x7B	{
GS1 DataBar Expanded	0x7D	}
GS1 DataBar Stacked	0x59	Y
2 of 5		
Interleaved 2 of 5	0x65	e
Matrix 2 of 5	0x6D	m
Industrial 2 of 5	0x66	f
MSI	0x67	g
UPC		
UPC-A	0x63	c
UPC-E0	0x45	E
UPC-E1	0x61	a
Aztec Code	0x41	A
Data Matrix	0x77	w
PDF417	0x72	r
Micro PDF417	0x52	R
QR	0x73	s
Micro QR	0x51	Q

Appendix 3 Serial Command Format

Name		Description
Head	2 byte	包头分为发送包头和反馈包头： 输入包头固定为：0x57 0x00 返回包头固定为：0x31 0x00
Device ID	1 byte	在非 RS-485 串口模式下，固定值为 0x00，在 RS-485 串口模式下该值可设置，用于多机通信中区分不同的设备。
Command Status	[7:4]	值为 0000 时，表示该指令为设置指令。
		值为 0001 时，表示该指令为获取状态指令。
	[3:0]	值为 0000 时，表示该指令正确动作。
		值为 0001 时，表示该指令错误动作。
Data Length	4 byte	当前指令中的数据段长度
Data	N byte	当数据长度为 0 byte 时，不需要填充该段内容。
CRC16 Check Digit	2 byte	除去 CRC16Check Digit 的所有数据进行 CRC16Check Digit。
Notice	1. 韦根通讯不支持指令控制！ 2. 指令控制不支持批处理！	

Appendix 4 Control Character

Dec	ASCII	Control + ASCII	Alt + Keypad
1	SOH (Start of Header)	Ctrl+A	Alt+001
2	STX (Start of Text)	Ctrl+B	Alt+002
3	ETX (End of Text)	Ctrl+C	Alt+003
4	EOT (End of Transmission)	Ctrl+D	Alt+004
5	ENQ (Enquiry)	Ctrl+E	Alt+005
6	ACK (Acknowledgment)	Ctrl+F	Alt+006
7	BEL (Bell)	Ctrl+G	Alt+007
8	BS (Backspace)	Back Space	Alt+008
9	HT (Horizontal Tab)	Tab	Alt+009
10	LF (Line Feed)	Ctrl+P	Alt+010
11	VT (Vertical Tab)	Ctrl+Q	Alt+011
12	FF (Form Feed)	Ctrl+R	Alt+012
13	CR (Carriage Return)	Enter	Alt+013
14	SO (Shift Out)	Ctrl+N	Alt+014
15	SI (Shift In)	Ctrl+O	Alt+015
16	DLE (Data Link Escape)	Ctrl+P	Alt+016
17	DC1 (XON) (Device Control 1)	Ctrl+Q	Alt+017
18	DC2 (Device Control 2)	Ctrl+R	Alt+018
19	DC3 (XOFF) (Device Control 3)	Ctrl+S	Alt+019
20	DC4 (Device Control 4)	Ctrl+T	Alt+020
21	NAK (Negative Acknowledgement)	Ctrl+U	Alt+021
22	SYN (Synchronous Idle)	Ctrl+V	Alt+022
23	ETB (End of Trans. Block)	Ctrl+W	Alt+023
24	CAN (Cancel)	Ctrl+X	Alt+024
25	EM (End of Medium)	Ctrl+Y	Alt+025
26	SUB (Substitute)	Ctrl+Z	Alt+026
27	ESC (Escape)	Ctrl+[	Alt+027
28	FS (File Separator)	Ctrl+\	Alt+028
29	GS (Group Separator)	Ctrl+]	Alt+029
30	RS (Request to Send)	Ctrl+^	Alt+030
31	US (Unit Separator)	Ctrl+_	Alt+031

Appendix 5 ASCII Table

Characters with a light yellow background are control characters, and characters with a white background are displayable characters.

Binary	Decimal	Hex	Char
0	0	0	NUL (NULL)
1	1	1	SOH (Start Of Heading)
10	2	2	STX (Start Of Text)
11	3	3	ETX (End Of Text)
100	4	4	EOT (End Of Transmission)
101	5	5	ENQ (Enquiry)
110	6	6	ACK (Acknowledge)
111	7	7	BEL (Bell)
1000	8	8	BS (Backspace)
1001	9	9	HT (Horizontal Tab)
1010	10	0A	LF/NL(Line Feed/New Line)
1011	11	0B	VT (Vertical Tab)
1100	12	0C	FF/NP (Form Feed/New Page)
1101	13	0D	CR (Carriage Return)
1110	14	0E	SO (Shift Out)
1111	15	0F	SI (Shift In)
10000	16	10	DLE (Data Link Escape)

10001	17	11	DC1/XON
			(Device Control 1/Transmission On)
10010	18	12	DC2 (Device Control 2)
10011	19	13	DC3/XOFF
			(Device Control 3/Transmission Off)
10100	20	14	DC4 (Device Control 4)
10101	21	15	NAK (Negative Acknowledge)
10110	22	16	SYN (Synchronous Idle)
10111	23	17	ETB (End of Transmission Block)
11000	24	18	CAN (Cancel)
11001	25	19	EM (End of Medium)
11010	26	1A	SUB (Substitute)
11011	27	1B	ESC (Escape)
11100	28	1C	FS (File Separator)
11101	29	1D	GS (Group Separator)
11110	30	1E	RS (Record Separator)
11111	31	1F	US (Unit Separator)
100000	32	20	(Space)
100001	33	21	!

100010	34	22	"
100011	35	23	#
100100	36	24	\$
100101	37	25	%
100110	38	26	&
100111	39	27	'
101000	40	28	(
101001	41	29	)
101010	42	2A	*
101011	43	2B	+
101100	44	2C	,
101101	45	2D	-
101110	46	2E	.
101111	47	2F	/
110000	48	30	0
110001	49	31	1
110010	50	32	2
110011	51	33	3
110100	52	34	4
110101	53	35	5
110110	54	36	6
110111	55	37	7

111000	56	38	8
111001	57	39	9
111010	58	3A	:
111011	59	3B	;
111100	60	3C	<
111101	61	3D	=
111110	62	3E	>
111111	63	3F	?
1000000	64	40	@
1000001	65	41	A
1000010	66	42	B
1000011	67	43	C
1000100	68	44	D
1000101	69	45	E
1000110	70	46	F
1000111	71	47	G
1001000	72	48	H
1001001	73	49	I
1001010	74	4A	J
1001011	75	4B	K
1001100	76	4C	L
1001101	77	4D	M

1001110	78	4E	N
1001111	79	4F	O
1010000	80	50	P
1010001	81	51	Q
1010010	82	52	R
1010011	83	53	S
1010100	84	54	T
1010101	85	55	U
1010110	86	56	V
1010111	87	57	W
1011000	88	58	X
1011001	89	59	Y
1011010	90	5A	Z
1011011	91	5B	[
1011100	92	5C	\
1011101	93	5D	]
1011110	94	5E	^
1011111	95	5F	_
1100000	96	60	`
1100001	97	61	a
1100010	98	62	b
1100011	99	63	c

1100100	100	64	d
1100101	101	65	e
1100110	102	66	f
1100111	103	67	g
1101000	104	68	h
1101001	105	69	i
1101010	106	6A	j
1101011	107	6B	k
1101100	108	6C	l
1101101	109	6D	m
1101110	110	6E	n
1101111	111	6F	o
1110000	112	70	p
1110001	113	71	q
1110010	114	72	r
1110011	115	73	s
1110100	116	74	t
1110101	117	75	u
1110110	118	76	v
1110111	119	77	w
1111000	120	78	x
1111001	121	79	y

1111010	122	7A	z
1111011	123	7B	{
1111100	124	7C	
1111101	125	7D	}
1111110	126	7E	~
1111111	127	7F	DEL (Delete)

